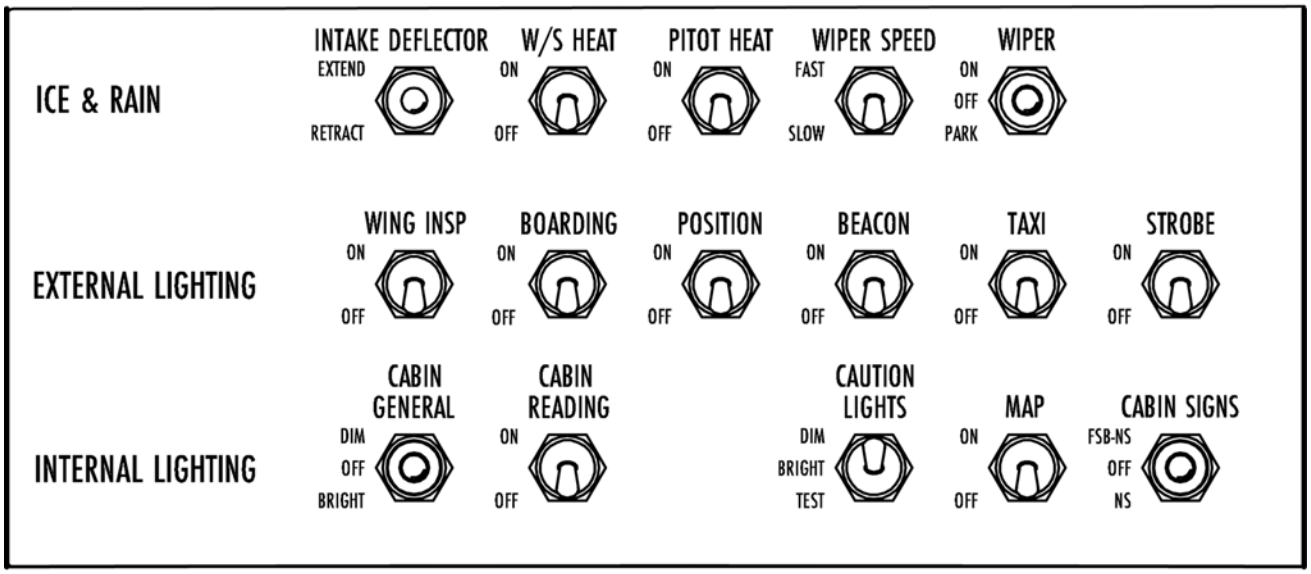
Flight Compartment Lights (Mod 6/2028) – General Data

1. General

The flight compartment lights include the dome light, aisle light on control column pedestal, pilot and co-pilot map lights, primary instrument panel lighting, secondary instrument panel flood lighting, foot well lighting and caution lighting.

The interior lighting system consists of six overhead general lights, twenty passenger reading lights, an entrance light, forward and aft baggage compartment lights, and cabin sign lights. All lights are LED (light-emitting diode).

All light switches are located on the middle row (for external lighting) or the bottom row (for internal lighting) of the lower left pilot sub-panel. The landing light switches are located on the overhead switch panel.



External and Internal Lighting Switches
Figure 1

2. Panel and Instrument Lighting

A. Panel Lighting

General instrument panel illumination is provided by LED arrays. Specific instruments and panels are lit internally or illuminated by LED post lights. For further detail refer to PSM 1-64-2W, Chapter 33.

B. Flight Compartment Dome and Aisle Lights

A flight compartment dome light is installed on the center aft overhead console and is supplied power from the hot battery bus. The dome light employs both clear and coloured LEDs to emit light at a colour temperature similar to daylight.

An additional aisle light is installed on the aft face of the control column pedestal, below the data entry keypad, for the purpose of illuminating the well below the two pilot seats.

Each of these lights have their own switches.

C. Flight Compartment Map Lights

Two flight compartment map lights are mounted on the center aft overhead console, just forward and to each side of the dome light. The map lights have a concentric control for dimming. A toggle switch labeled MAP, controlling master power to both map lights is provided on the left pilot sub-panel.

D. ESIS Bezel Lighting

The ESIS bezel light is dimmable through R2 INSTRUMENT dimmer control when power is on the aircraft, and the ESIS power switch is set to ARM. When power is removed from aircraft, and ESIS power switch is set to ARM, the ESIS power relay in the junction box is disengaged which causes the ESIS bezel light to remain at full brightness, and cannot be dimmed.

E. Control and Electrical Supply

1) Panel Lights

The LPS 99–1128 (PS1) power supply supplies 5V and 28V variable voltage to panel lights, avionics instruments, as well as the static valve lighting, and standby compass lighting. The unit is powered by a five amp breaker on the left bus, and dimming is controlled by a 10k Ω rheostat on the Control Column Dimming Panel. The PS1 supplies 0-5 volt variable voltage to the Fuel Selector Panel overlay, the Overhead Switch Panel overlay, the Left Hand Panel overlay, the Right Hand Panel overlay, the Cockpit Dimmer Panel overlay, and TB13. It supplies 0-28V variable voltage to avionics instruments, the standby compass, and TB13. It also supplies 0-28V variable voltage to the trim panel overlay, the flap selector post lights, and the static valve illumination through switch S107 TRIM PANEL LIGHT switch. The PS1 system also includes a 9-pin maintenance port, which is utilized by any PC computer equipped with an RS422 adapter unit maintenance or set up. The display revisionary panel is supplied by the 0-28V output, through R101, a 150 Ω drop resistor, so as to supply 14V max to panel.

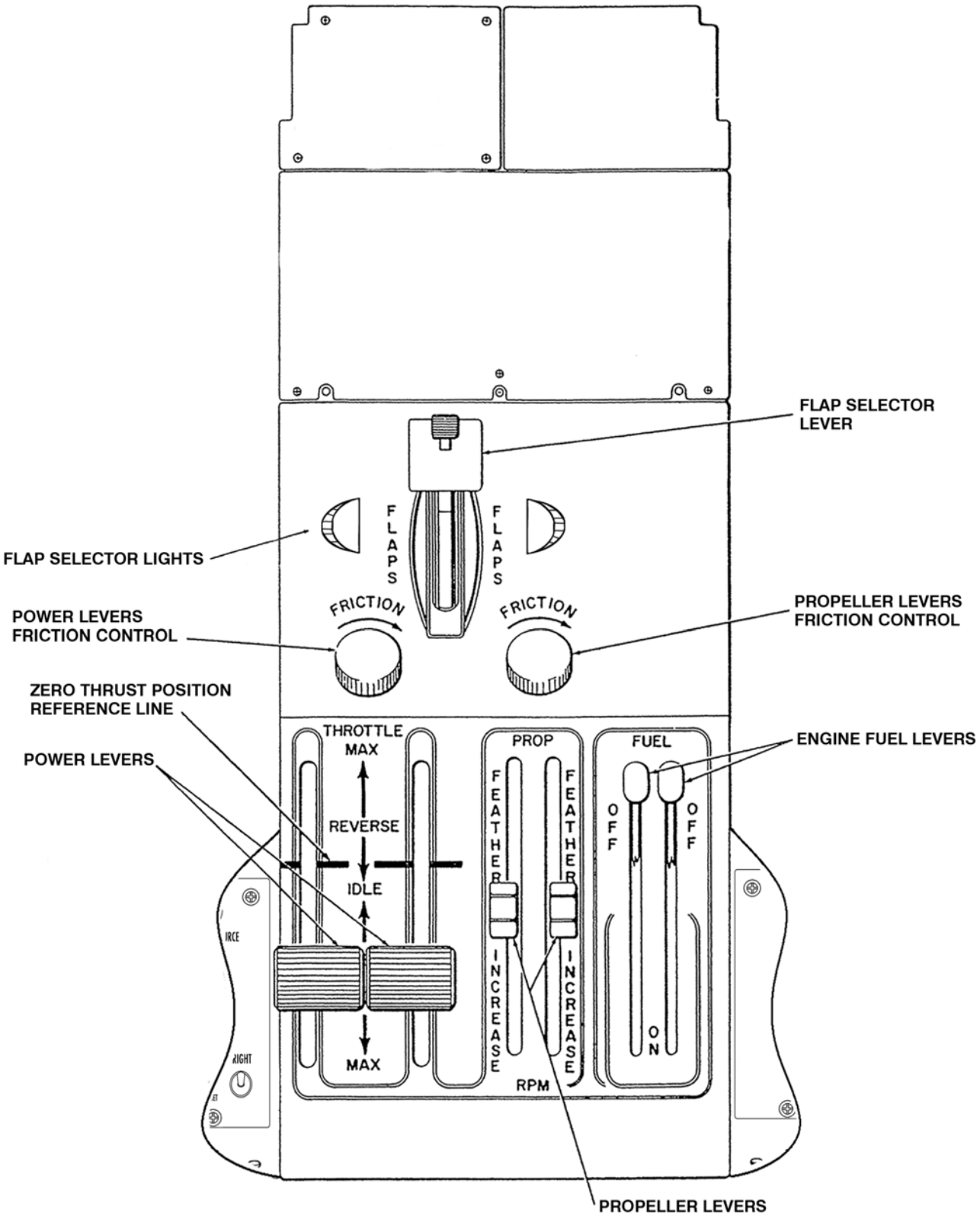
2) Flood/Thunderstorm Lights

The LT–50 () or LT–50(D) (PS2) power supply supplies 0-28V variable voltage to the avionic c/b panel illumination, eye-height reference lighting, foot lights, and instrument panel flood lights. The unit is powered by a five amp breaker on the left bus, the aisle light is powered by the same circuit breaker however, it is dimmed by its own switch (S40) AISLE LIGHT switch. Dimming of PS2 is controlled by R1 FLOOD, a 5k Ω rheostat on the Control Column Dimming Panel. Switch (S42) FOOTWELL activates foot lights and avionic circuit breaker panel illumination. Instrument panel flood lights, eyeheight reference device is activated with switch S41 OFF EYE REFERENCE light located on the window post.

3. Caution Lights Dimming/Test Switch

The Caution Lights Dimming Control Box forms part of the master warning and master caution system, and consists of a remote mounted light dimmer control box which is capable of controlling up to 12 independent lamps with full test functions.

A three-position switch labelled CAUTION LIGHTS with positions DIM, BRIGHT, and TEST is located on the lower left sub-panel. The momentary TEST position of the switch permits a simultaneous check of Autofeather, Master Caution, Master Warning, Stall, and the Fire Caution Lights. The Fire lights can also be tested with their own dedicated switch. The test and intensity circuit is protected by a CAUTION LTS NORMAL circuit breaker on the avionics circuit breaker panel. The DIM and BRIGHT positions provide alternative levels of lighting brilliance for the MASTER CAUTION and MASTER WARNING lights, the autofeather system SELECT and ARMED indicator lights, and the stall warning light, and the engine FIRE WARNING lights. In the event of failure of the control system used for caution light dimming, a reversionary, non-dimming mode can be selected by moving the CAUTION LIGHTS DIMMING switch on the lower right sub-panel from NORMAL to BYPASS. It is powered by the CAUTION LTS BYPASS circuit breaker on the avionics circuit breaker panel.



Overhead Console – Typical
Figure 2

4. Adjustment/Test

A. Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Pull the following circuit breakers:

1) MAP LIGHTS located on the main circuit breaker panel.

2) CLOCK AND DOME LIGHT on the rear circuit breaker panel.

3) PANEL LIGHTS located on the avionics circuit breaker panel.

4) FLOOD LIGHTS located on the avionics circuit breaker panel.

C. Test Map Lights

1) Reset the MAP LIGHTS circuit breaker in the main circuit breaker panel.

2) Select the MAP switch to ON.

3) Pilot MAP Light

a) Select the dimmer to MAX and verify the lamp illuminates.

b) Cycle the MAP switch to the ON and OFF positions. Verify the lamp goes ON/OFF with the switch position.

c) Operate across dimming range. Verify the brightness of lamp varies, smooth operation and no flickering.

d) Dimmer to OFF. Verify the lamp extinguishes.

4) Repeat Para 4.C.3) for the Co-pilot MAP light.

5) Select MAP switch to OFF.

D. Test Dome Light

1) Check that MASTER and POWER SOURCE switches have no effect on entrance lights operation.

Note Will not operate on External Power. Battery must be connected. Light is on the Hot Battery Bus.

2) Reset the CLOCK AND DOME LIGHT circuit breaker in the rear circuit breaker panel.

3) Operate the Dome Light switch on. Verify the lamp illuminates.

4) Operate the Dome Light switch off. Verify the lamp extinguishes.

E. Test Panel Lights

1) Select MASTER switch to OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches to OFF (Post Mod 6/2314).

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Reset the PANEL LIGHTS circuit breakers on the avionics circuit breaker panel.

4) Select the INSTRUMENT dimmer to full bright (CW).

5) Verify the following lights are illuminated and illumination across the panel is adequately balanced:

- HF Control (if installed)
- Audio Panels (two)
- Flight Controller Panel
- PFD Control Panels (two)
- Display Bezels (four)
- ESIS Bezels (if installed)
- Compass
- Clock
- MFD Control Panel (one in pedestal)
- Reversionary Control Panel
- Dimmer Panel
- Fuel Selector Panel
- Overhead Panel
- LH Switch Panel
- RH Switch Panel

6) Select TRIM PANEL LT switch to ON.

7) Verify that the Trim Panel, Flap Selector post lights and the Static Valve post lights are illuminated.

8) Operate the INSTRUMENT dimmer across the dimming range. Verify the brightness of lamp varies, smooth operation and no flickering.

9) Cycle the TRIM PANEL LT switch ON/OFF. Verify the TRIM PANEL, FLAP SELECTOR post lights and the STATIC VALVE post lights illuminate accordingly.

10) Select the INSTRUMENT dimmer to dim (CCW) position. Verify the TRIM PANEL, FLAP SELECTOR post lights and the STATIC VALVE post lights dim.

11) Select TRIM PANEL LT switch to OFF.

F. Test Flood Lights

1) Reset FLOOD LIGHTS circuit breaker in the avionics circuit breaker panel.

2) Select the AISLE LT switch (Pedestal) to BRT (up). Verify the AISLE LT is bright.

3) Select the AISLE LT switch (Pedestal) to DIM (down). Verify the AISLE LT is dim.

4) Select the AISLE LT switch (Pedestal) to OFF (center). Verify the AISLE LT is extinguished.

5) Select the FLOOD light dimmer to full bright position.

6) Verify the instrument panel visor lights, six lights, are illuminated.

7) Select the FOOTWELL switch to ON/OFF. Verify the footwell lights (two) and the avionics circuit breaker post lights cycle on/off with switch position.

8) Select the FOOTWELL switch to ON.

9) Select the EYE REFERENCE LIGHT switch to ON/OFF. Verify the eye reference light cycles with the switch position.

10) Select the EYE REFERENCE LIGHT switch to ON.

11) Operate the FLOOD dimmer across the dimming range. Verify the brightness of Instrument Panel Visor, Footwell, Avionics Circuit Breaker Panel Post and Eye Reference Lamps varies, smooth operation and no flickering.

12) Select FOOTWELL switch to OFF.

13) Select the EYE REFERENCE LIGHT switch to OFF.

14) Select the FLOOD dimmer to full dim.

15) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

If there is a problem with the Panel or Footwell lights dimming function, please contact Viking Air Ltd., technical services.

G. Test Panel Light Power Supply

Note When first installing the LPS 99–1128, follow the manufacturer's installation instructions before testing the system. Without proper set-up with a computer through the maintenance port, the power supply will have the factory settings for the variable dimming.

Operating the LPS 99–1128 with factory settings and no configuration tables loaded will not damage any of the lighting fixtures. When operated with factory settings, the illumination balance will be poor across the instrument panel, most naturally, the clock and the reversionary panel will be noticeably brighter than after equipment when the LPS is set between medium and high settings.

The clock, compass and eyeheight trim panel have incandescent lamps and may appear to have a warmer colour than other cockpit lights at some brightness settings.

1) With power on bus, and CB277 PANEL LIGHTS circuit breakers pushed in, rotate R2 rheostat labeled INSTRUMENT on the control column dimming panel to bright. Ensure the fuel selector panel, overhead switch panel, left hand panel, right hand panel, standby compass, and avionics instruments are at full brightness.

2) Activate switch S107 TRIM PANEL LIGHT switch, to the ON position. Ensure the flap selector post lights, static valve illumination, and trim panel overlay illuminate at their brightest

3) Rotate R2 rheostat to dim. Ensure the fuel selector panel, overhead switch panel, left hand panel, right hand panel, standby compass, avionics instruments, flap selector post lights, static valve illumination, and trim panel overlay dim.

4) Reset all switches and controls to appropriate settings.

H. Test Flood/Thunderstorm Light Power Source

1) With power on bus, and CB279 FLOOD LIGHTS circuit breaker pushed in, rotate R1 rheostat labeled FLOOD on the control column dimming panel to bright. Ensure instrument panel flood lights are at full brightness.

2) Activate switch S42 FOOTWELL to the ON position. Ensure the foot lights, and avionic circuit breaker panel illuminations are at full brightness.

3) Activate switch S41 OFF EYE REFERENCE light to the ON position. Ensure the eyeheight reference light is at full brightness.

4) Rotate R1 rheostat to dim. Ensure the instrument panel flood lights, eyeheight reference light, foot lights, and avionic circuit breaker panel lighting has dimmed.

5) Activate S40 AISLE LIGHT switch to the BRIGHT position. Ensure the aisle light is at full brightness.

6) Activate S40 AISLE LIGHT switch to the DIM position. Ensure the aisle light has dimmed.

7) Reset all switches and controls to appropriate settings.

I. ESIS Bezel Light Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Set ESIS POWER switch to ARM.

3) Rotate R2 INSTRUMENT dimmer on control column dimming panel to full bright. Ensure bezel is at full brightness.

4) Rotate R2 INSTRUMENT dimmer control to fully dim. Ensure bezel fully dims.

5) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

6) Ensure ESIS bezel light is at full brightness, and does not dim when R2 INSTRUMENT dimmer control is rotated.

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